Given parameter:

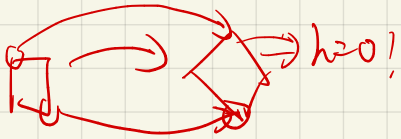
{ poly-degree ⟶ need this to construct basis
1D quadrature order
cfl:
{ ofdg
charac-decompos
{ positive-limiter
density- 10^{-13}
pressure- 10^{-13}
{ max try
max steps
tol $= 10^{-13}$

> options

Basis   shu. pq   Given $\bar{x}, \bar{y}$, h_x, h_y, degree.

$$h_x = x_{i+\frac{1}{2}, j+\frac{1}{2}} - x_{i-\frac{1}{2}, j+\frac{1}{2}}$$

Is not good is cell



So I change pq's h_x

as $h_x = X_{max} - X_{min}$ $h_y = Y_{max} - Y_{min}$

Let $\xi = 2\frac{x-\bar{x}}{h_x}$ $\eta = 2\frac{y-\bar{y}}{h_y}$

$\phi_0^{is} = 1$ $\phi_1 = 2\frac{x-\bar{x}}{h_x} = \xi$ $\phi_2 = 2\frac{y-\bar{y}}{h_y} = \eta$

Actually we can make it more generous raw Basis

$$\underbrace{1}_{0} \quad \underbrace{\xi \quad \eta}_{1} \quad \underbrace{\frac{1}{2}\xi^2 \quad \frac{1}{2}\eta^2 \quad \xi\eta}_{2} \quad \underbrace{\xi^3 \quad \xi^2\eta^1 \quad \xi\eta^2 \quad \eta^3}_{3} \quad \ldots$$

Abstract as $(0,0) \mid (1,0)\ (0,1) \mid (2,0)\ (0,2)\ (1,1) \mid (3,0)\ (0,3)\ (2,1)\ (1,2))$

$(4,0)\ (0,4)$
$(3,1)\ (1,3)$
$(2,2)$

as basis_powers

We just play degree 0~3 this time.

input (x,y)

$0\ 0\ (1,0)\ (0,1)$ - - - — —

output $(1, \xi, \eta, \frac{1}{2}\xi^2, \cdots\cdots)$.

Given $\dfrac{d^{n+m} \xi^p \eta^q}{dx^n\, dy^m} = \dfrac{d^n \xi^p}{dx^n} \cdot \dfrac{d^m \eta^q}{dy^m} = p(p-1)\cdots(p-n+1)\, \xi^{p-n} (\xi_x)^n$

$q(q-1)\cdots(q-m+1)\, \eta^{q-m} (\eta_y)^n$

$\xi_x = \left(\frac{2}{h_x}\right) \quad \eta_y = \left(\frac{2}{h_y}\right)$

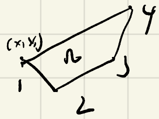
$$\boxed{\frac{p!}{(p-n)!}\, \xi^{p-n}\, \frac{q!}{(q-m)!}\, \eta^{q-m} \left(\frac{2}{h_x}\right)^n \left(\frac{2}{h_y}\right)^m}$$

According to basis, for each cell we need to compute

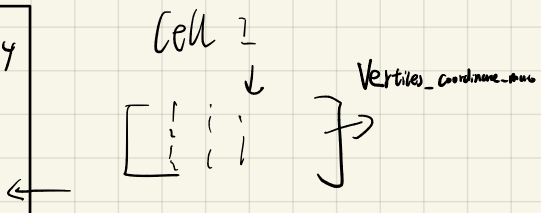
h_x h_y, $\bar{x}$, $\bar{y}$;

To get this we need

$$\begin{bmatrix} x_1 & \cdots & x_4 \\ y_1 & \cdots & y_4 \end{bmatrix}$$



① vertex_coordinate_mat



to calculate $\bar{x} = \iint_{\Omega} x \cdot dx\,dy = \sum_i x(q_i)\, w_i$ as we showed in quadrature

We can save $\boxed{q_i}$ ② $\boxed{w_i}$ ③ for each cell

Then we also can get $\text{area} = \sum 1 w_i$ ④

$$\bar{x} = \frac{\sum x(q_i) w_i}{\text{area}} \Rightarrow (\bar{x}, \bar{y}) \text{ centroid } ⑤$$

$$\bar{y} = \frac{\sum y(q_i) w_i}{\text{area}},$$

⑥ — $V_{x_{min}}$ $V_{x_{max}}$

$V_{y_{min}}$ $V_{y_{max}}$

$\Downarrow$

⑦ h_x ⑧ h_y

$= V_{x_{max}}[1] - V_{x_{min}}[1]$ . . .

Then we want orthogonal basis for all ϕ ...

$$\tilde{\phi}(x) = \phi(x) - \frac{\int \phi \cdot 1}{\int 1} = \phi(x) - \frac{\sum \phi(q_i) w_i}{\text{area}}$$

because $\int \tilde{\phi} \cdot 1 = \int \phi(x) \cdot 1 - \int \frac{\left(\int \phi \, 1 \, dx\right)}{\int 1} dx$

$$= \int \phi \, dx - \int \phi \cdot 1 \frac{\int 1 d}{\int 1} = 0$$

So we will use $\boxed{\frac{\int \phi(q_i) w_i}{\text{area}}}$ — raw_mean to compute $\tilde{\phi}(x)$

We compute $\begin{bmatrix} \phi_1(q_1) & \cdots & \phi_1(q_n) \\ \vdots & & \vdots \\ \phi_m(q_1) & \cdots & \phi_m(q_n) \end{bmatrix}$ as raw_basis

raw_basis

basi_dx $\begin{bmatrix} \phi_{x_1}(q_1) & \cdots & \phi_{x_1}(q_n) \\ \phi_{x_m}(q_1) & & \phi_{x_m}(q_n) \end{bmatrix}$

basis_dy $\Big[$

$$\text{raw\_basis}\underbrace{\begin{bmatrix} \quad & \quad \\ & \\ & \end{bmatrix}}_{\text{area}}\begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} = \text{raw\_mean}$$

$$\text{basis} = \begin{bmatrix} \text{raw\_basis} \\ \\ \\ \end{bmatrix} - \begin{bmatrix} \text{raw\_mean} \\ \vdots \\ \text{raw\_mean}_n \end{bmatrix}$$

$$\qquad\qquad\downarrow \qquad\qquad\qquad\qquad \downarrow$$

Save ⑩ ⑨

basis_dx ⑪ basis_dy ⑫

This is better than (2.15)

Let's compute Mass_matrix here:

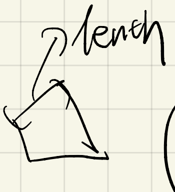
$$M_{ij} = \int_\Omega \phi_i \phi_j \, dx\,dy = \sum_i \phi_i(q_i)\phi_j(q_i) w_i$$

$$\vec{\phi}(q_i) \otimes \underline{\vec{\phi}(q_i)^T}$$

$$M = \sum w_i \left| \begin{array}{ccc} \phi_k(q_i)\phi_l(q_i) & \cdots & \text{— } k\text{th row} \\ \vdots & & \vdots \\ \cdots & \cdots & n_b \times n_b \\ l\text{-col} & & \end{array} \right.$$

⑬

final record



⑭ max lenth
min lenth
(because we need it to $\boxed{\text{OF}}$)

Loops all cell ↑ buildit

We have $\underline{\text{Cells\_ geometries}}\left[\overset{1}{\underset{\substack{\downarrow\\ \text{geo}}}{0}}\ \overset{\substack{\text{cell-indep}\\ \downarrow}}{2}, \cdots\cdots\right]$

Which is time dependence!

Function eval-basis.

Now given geo of a cell

eval basis at x in cell

$$\vec{\phi}(x) = \vec{\phi}_{raw}(x) - \overrightarrow{\text{geo.row\_mean}}.$$

$$\partial\vec{\phi}(x) = \overrightarrow{\partial\phi_{raw}(x)}$$

Now with geometry, we can do more:

Function eval-solution with coefficient of basis:

$$\text{Given } \underset{\substack{\downarrow \\ \text{Cell k}}}{C_k} = \begin{array}{cccccc} & \downarrow 1 & 2 & 3 & 4 & \\ \text{Number of basis}\left\{\begin{matrix} \\ \\ \\ \end{matrix}\right. & \left[\begin{matrix} C_{11} \\ \vdots \\ C_{nb1} \end{matrix}\right. & & & \left.\begin{matrix} C_{14} \\ \\ \\ \end{matrix}\right] & \\ & U_1 & & & U_4 & \end{array} = C[:\,:\,k]$$

$$\left(\begin{pmatrix} u_1 \\ \vdots \\ u_4 \end{pmatrix} \text{is conservation vector}\right)$$

C'

$$u_k = [:\,,k]$$

$$U_j(x) = \sum_i C_{ij}\phi_j \qquad \frac{dU_j(x)}{dx} = \sum_i C_{ij}\frac{d\phi_{ij}}{dx}$$

$$V_k = \begin{bmatrix} u_1 \\ \vdots \\ u_4 \end{bmatrix}(x) = \begin{matrix} \sum C_{i1}\phi_i \\ \vdots \\ \sum C_{i4}\phi_i \end{matrix} = \begin{bmatrix} C_{11} & \cdots & C_{nb1} \\ & & \\ & & \end{bmatrix}\begin{bmatrix} \phi_1 \\ \vdots \\ \phi_{nb} \end{bmatrix} = C_k^T \vec{\phi}(x)$$

$$\boxed{\text{Note } \quad C_{11} \cdots C_{14} \text{ is } \bar{U}}$$

Function def: moment

$$\underset{\substack{\downarrow \\ \text{Cell k}}}{Q_k} = \begin{bmatrix} \int\phi_1 u_1 & \cdots & \int\phi_1 u_4 \\ \int\phi_2 u_1 & \cdots & \vdots \\ \vdots & \ddots & \vdots \\ \underbrace{\int\phi_{nb} u_1}_{M \text{ of } u_1} & \cdots & \underbrace{\int\phi_{nb} u_4}_{M \text{ of } u_4} \end{bmatrix} = Q[:\,:\,,k]$$

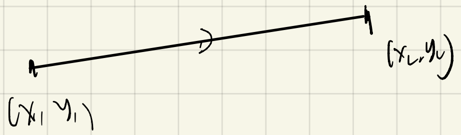
$$Q_{kij} = \int\phi_i U_j = \int\phi_i \sum_k C^T_{jk}\phi_k = \sum_k C^T_{jk}\int\phi_i\phi_k = \sum_k C^T_{jk} M_{ik}$$

$$= \sum_k M_{ik} C_{kj}$$

$$\text{So } \begin{cases} Q_k = M_k C_k \\ C = M_k^{-1} Q_k = M_k \backslash Q_k \end{cases}$$

edge

quadrature (0,1)



$$\alpha\left[(x_2, y_2) - (x_1, y_1)\right] + (x_1, y_1)$$

$$(1-\alpha)(\bar{x}_1) + \alpha\,\bar{x}_2$$